Introduction to Data Science

Class-5

Association Analysis

- Association analysis discovers relationships or co-occurrences among items in a dataset.It is commonly used in market basket analysis.
- Example:
- Customers who buy **bread** often buy **milk**.

Correlation Analysis

- Correlation analysis identifies whether two or more variables are related.
- Suppose we observe:
- Study hours increase → Exam scores tend to increase.
- There may be a positive correlation.

Classification

- **Classification** is the process of finding a model that assigns data objects to predefined classes or categories.
- **Example**
- Predict whether an email is:
- **Spam**
- **Not Spam**
- **Classification Algorithms**
- Decision Tree
- Naïve Bayes
- k-Nearest Neighbors (k-NN)
- Support Vector Machine (SVM)
- Logistic Regression
- Random Forest
- Neural Networks

Prediction

- Prediction is used to predict a future or unknown value.
- Unlike classification, prediction often deals with a continuous numerical value.

Clustering Analysis

- **Clustering** groups similar data objects together without predefined class labels.
- It is an **unsupervised learning** technique.
- **Example**
- A supermarket has customer data but does not know the customer groups.
- A clustering algorithm may identify:
- **Group 1**
- Frequent high-spending customers
- **Group 2**
- Frequent low-spending customers

Outlier Analysis

- An **outlier** is a data object that differs significantly from the general pattern of the data.
- Outlier analysis identifies these unusual observations.
- ***Applications of Outlier Analysis***
- Outlier detection is useful for:
- Fraud detection
- Network intrusion detection
- Manufacturing defects
- Medical abnormality detection
- Financial risk analysis
- Sensor failure detection

Statistics

- Statistics is the study of the collection, organization,analysis, interpretation, and presentation of data. No statistics are possible without data.
- **Types of Statistics**
- **Descriptive Statistics:** The data is described by descriptive statistics**.**
- **Inferential Statistics:** Inferential statistics help in decision making by examining (Test of hypothesis) the project.

Types of Data used in statistics

- The data used in statistics can be of two types
- **Primary Data** - When a statistician collects data by himself or through his own team it is called primary data.
- **Secondary Data** - Secondary data are data collected from any other source. Secondary data is not collected by the statistician himself, but from any other statistic or source. Secondary data source is always to be mentioned

Statistical Method



Sample from Population

- The sample must be representative, that is, the sample must represent the whole population.

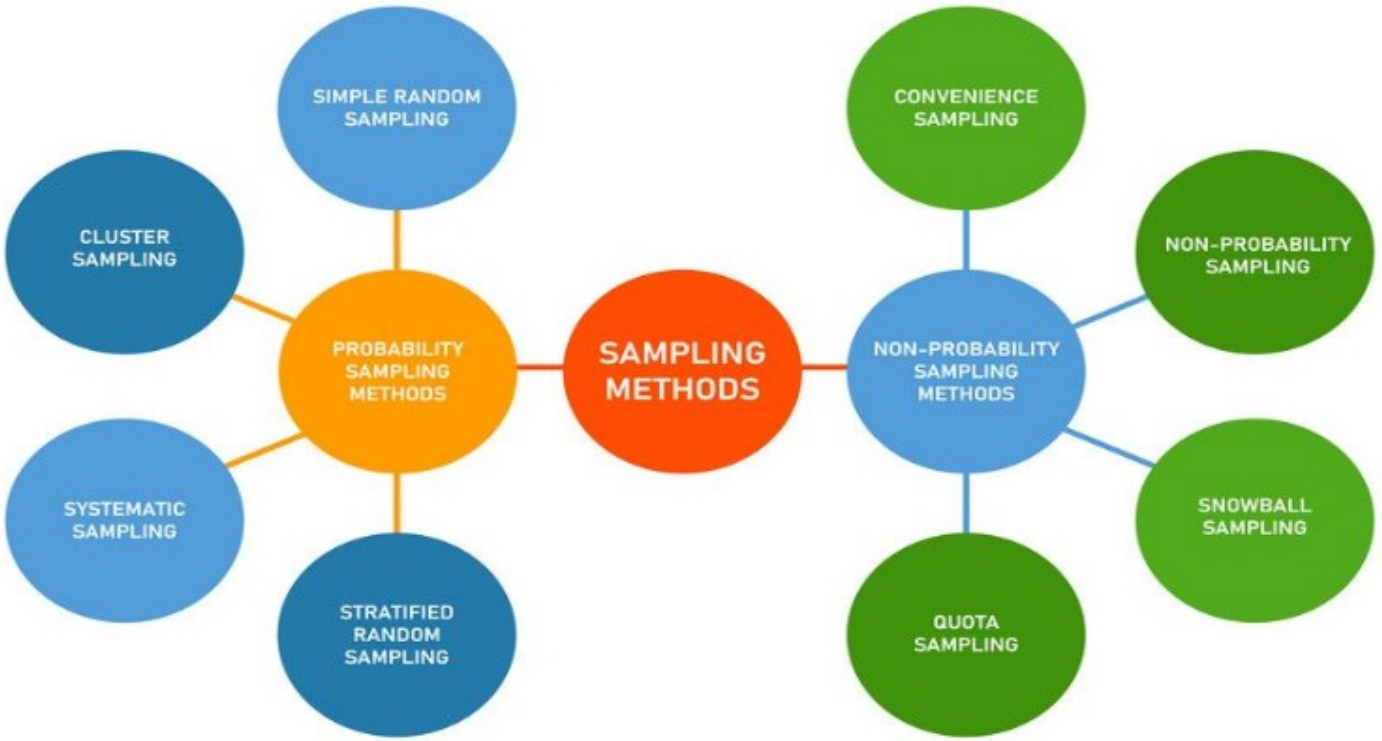
Example: If an ordinary person is asked to pick a mango from a basket, he will only choose a big, fresh and ripe one. As a result, the collected specimens became one-sided, meaning that not all of these specimen represent different types of mango. So the sample has to be selected in such a way that there is no chance of bias.

- If we do not select the correct sample then the results of the study will be wrong, so sampling is considered important in any study

Necessity of Sampling

- Suppose you want to know the average salary of all the IT professionals in Bangladesh, that is, all the IT professionals in Bangladesh are the population.
- In that case you have to go to all the IT professionals in Bangladesh and find out their salary and the average that will be obtained from the data obtained will be the actual average.
- When the data of all the members of the population is taken it is called census survey.
- In fact, in most cases it is not possible to collect data from all members of the population. Because it requires a lot of time, manpower and money. That is why a small part representing the population is taken instead of the whole population.

Types of Sampling



Probability and Non-probability Sampling

- **Probability Sampling** - Probability Sampling has the probability of being selected as a sample of each member of the population.

- **Non-Probability Sampling** - This method is called non-probability sampling because the researcher's judgment is important in selecting samples in non-probability sampling. Non-probability sampling usually does not use probability mechanism. The decision of the researcher is the rule in this case.

Types of Probability Sampling

- **Simple Random Sampling**
- **Cluster Sampling**

Simple Random Sampling

- **Simple Random Sampling** - The most fundamental method of probability sampling is simple random sampling. In this method a certain number of samples are selected from the population through probability mechanism.
- For example lottery ticket. Each ticket in the basket is likely to be selected at the time of ticket selection. Simple random sampling can also be done through random number generation.

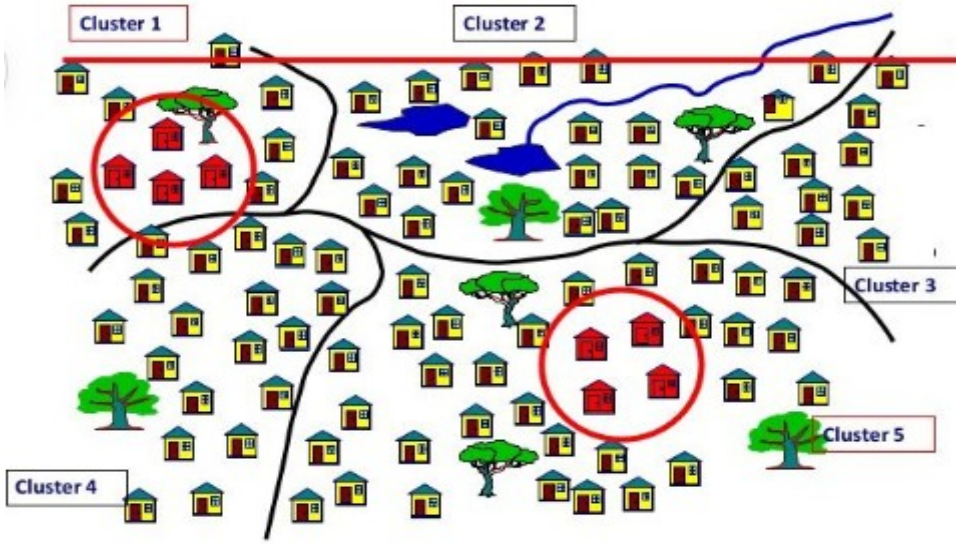




Cluster Sampling

- **Cluster sampling-** Cluster sampling is called geographical location based sampling. For example, if a district-upazila union based sample is taken in Bangladesh, it will be a cluster sample.
- Cluster samples can also be taken from other geographical divisions.

Cluster sampling can again be divided into two parts, one stage cluster and two stage cluster



One and two stage cluster sampling

- When all the components inside a cluster are taken as samples it is a one stage cluster. Suppose each ward of Dhaka city is considered as a cluster, in which case samples of all the people of the ward will be taken in one stage cluster.
- In the case of two stage clustering, instead of taking everyone inside a

cluster as a sample, a certain number of samples are selected by following a simple random sampling method. In other words, we want to take data of 100 people from each cluster so 100 people will be selected again from each cluster through simple random sampling.

Stratified Sampling

- **Stratified Sampling** - Stratified sampling is a homogenous group based sampling method. All members of the same group usually have the same characteristics, such groups are called strata.
- For example, we want to find out the average age of students in a class. This is why we can create two separate groups of female students and male students, so that it is possible to take separate samples from each homogenous group and each sample represents a group whose members are all homogenous or possess the same characteristics.

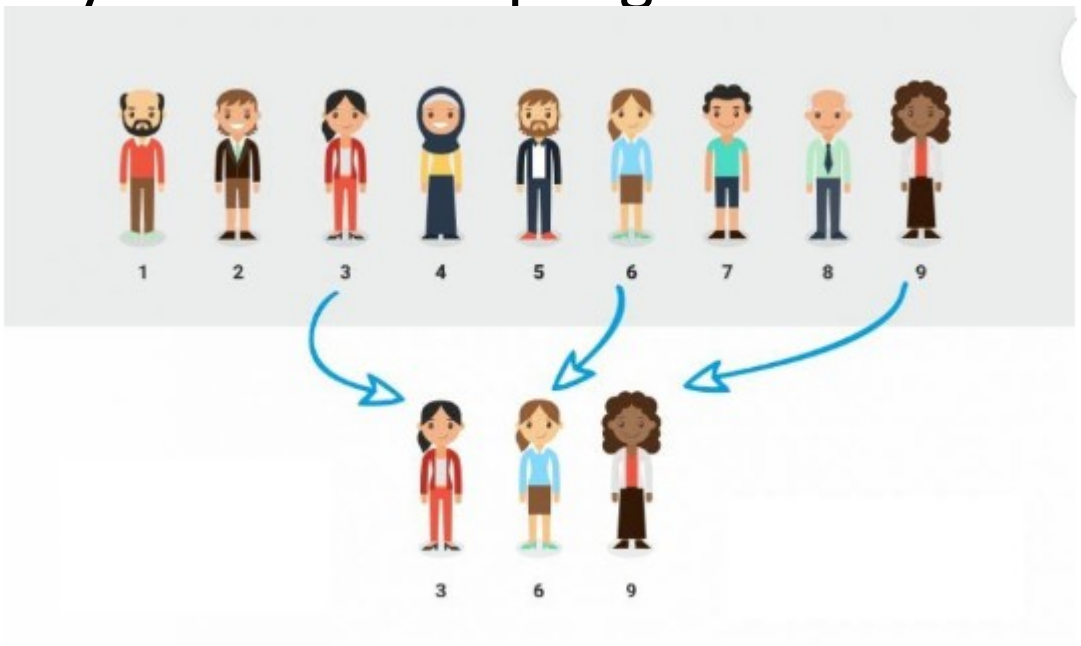




Systematic Sampling

- **Systematic Sampling** - Systematic sampling follows a specific

sequence. For example, every 2, 3, 4 or any number of data is taken. For example, if a researcher wants to study the number of family members living in a ward of Dhaka city, he wants to collect data of 100 families through systematic sampling.

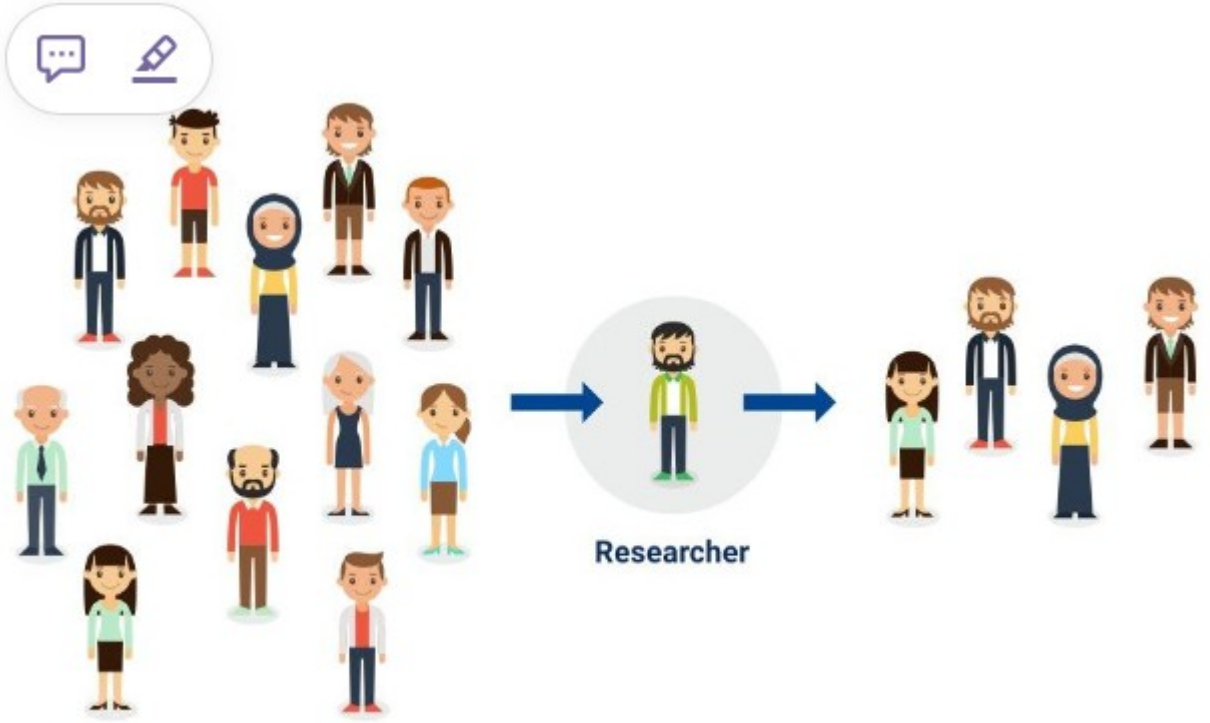


Types of Non-probability Sampling

- Judgmental Sampling
- Snow-Ball Sampling
- Convenience Sampling
- Quota Sampling

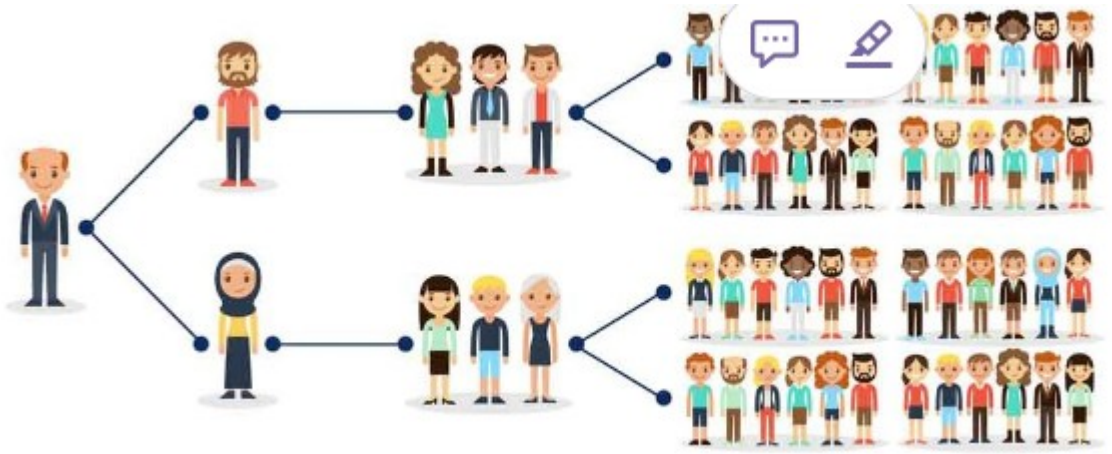
Judgmental Sampling

Judgmental Sampling - In judgmental sampling, the researcher selects samples from the population according to his own discretion.



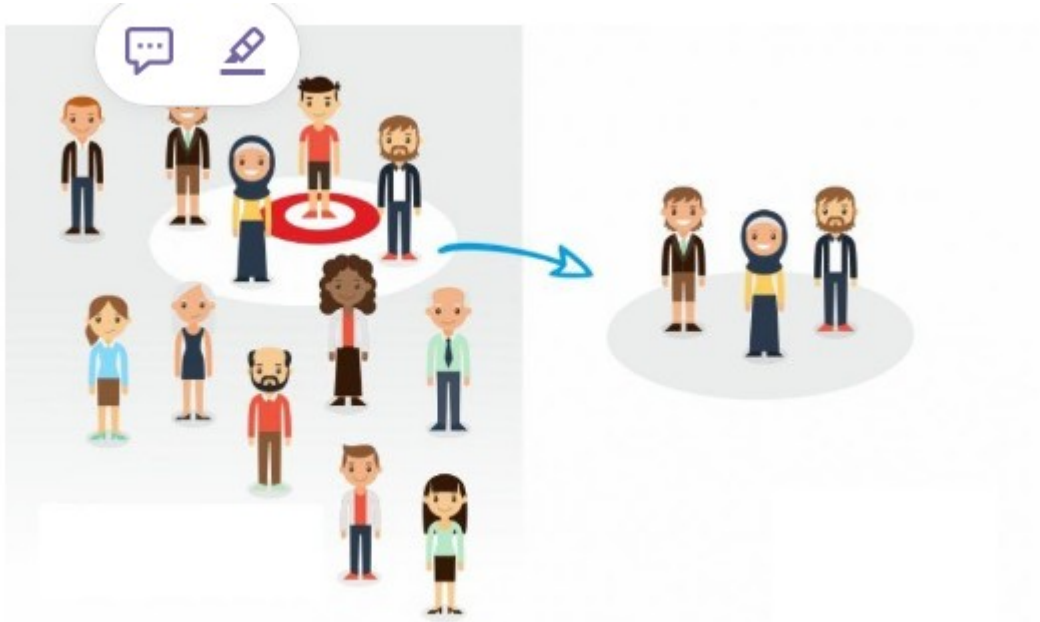
Snow-Ball Sampling

- Snow-Ball Sampling - It usually takes samples that are difficult to reach in parts of the population. In this case a sample serves as a reference to the next sample.
- **For example**, it is naturally difficult to find HIV patients or drug addicts because in most cases such people hide their identities or find them difficult to find. In such cases samples are collected by means of snow-ball sampling, in which case the reference is passed from one person to another, thus gradually reaching the entire population.



Convenience Sampling

- **Convenience Sampling** – It allows the researcher to select a member from his or her hand or a relatively readily available sample.
- **For example**, if a company wants to give feedback to customers about a product that is newly marketed, then feedback is taken from the buyers who are available in the market.



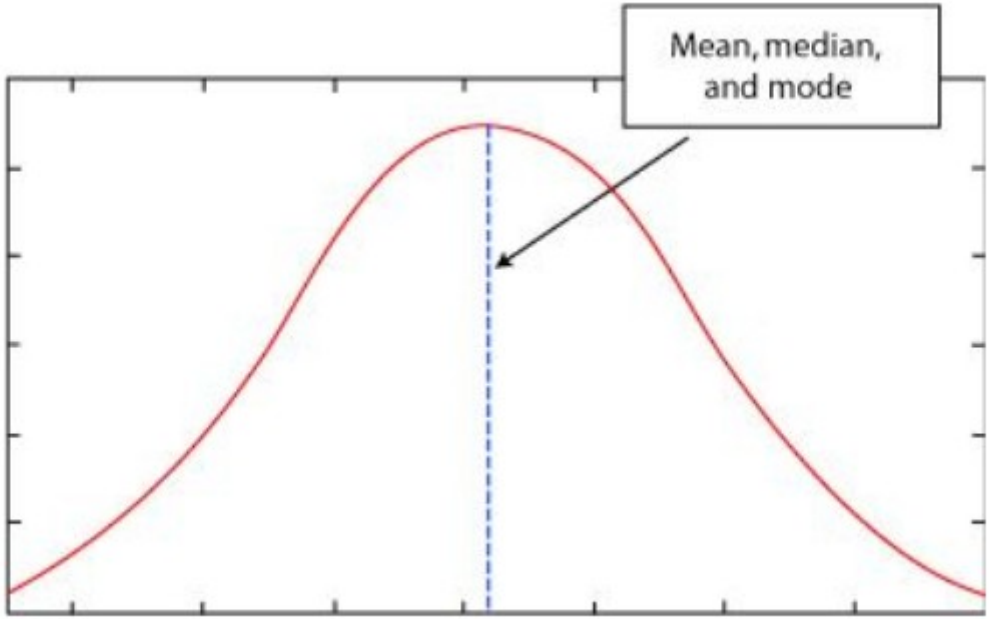
Quota Sampling

- **Quota Sampling** - In the case of quota sampling, usually a certain proportion or quota is followed in the sample selection.
- **For example**, if 80% of the population is male and 40% female, then the same ratio will be followed during sample selection.



Central Tendency

- Mean, mode and median is a measure of the central tendency of any data.
- Many of us look at the shape of a mountain peak, sand dune or bell shape when we format any data. Most of the data is concentrated in the middle. This tendencv to have data in the middle is called central tendency.



Mean

- Mean - Mean is average. In simple words, the sum of all the numbers is divided by the total number of numbers.
- Suppose the age of 10 students in a classis 14, 13, 11, 14, 15, 12, 16, 17, 11,14 respectively.

Data

14	13	11	14	15	12	16	17	11	14

$$\text{Mean } \bar{x} = \frac{\sum x}{n} = \frac{14+13+11+14+15+12+16+17+11+14}{10} = 13.7$$

That means we can say that the average age of students in that class is 13.7 years.

Mode

- **Mode-** Mode is the most common number or the number that comes up the most times.

Data									
14	13	11	14	15	12	16	17	11	14

Data	Frequency
14	3
13	1
11	2
15	1
12	1
16	1
17	1

Number 14 comes up most (3 times).
So, Mode=14

Median

- **Median** - Median is called the middle value. Median is the middle value that is obtained by sorting the data. The median value is easily obtained if the total observation of the data is an odd number, but if the observation is an even number then the average of any two data in the middle is taken as median.
  - If the no. of observations are odd, then $(\frac{n}{2}+1)th$= 5th data= 14
  - If the no. of observations are even, then $\frac{\frac{n}{2}th+\left(\frac{n}{2}+1\right)th}{2} = \frac{14+14}{2} = 14$

Data

14	13	11	14	15	12	16	17	11

Sorted Data

11	11	12	13	14	14	15	16	17